

ALIBABA CLOUD AI HACKATHON · HOSTED BY AL-KHIDMAT FOUNDATION

Mustahiq AI

Finding, verifying and fairly allocating aid — **and helping beneficiaries grow.**

A staff decision-support layer for Al-Khidmat's beneficiary programmes, plus a no-fee marketplace for micro-entrepreneurs.

A team submission · built end to end on free-tier infrastructure

Deciding who gets limited help is manual — and hard to defend



A household often qualifies for several programmes

but a field officer only has time to check one or two, so real needs are missed.



“Why them and not me?” has no clear answer

prioritisation lives in spreadsheets and memory; two officers rank the same people differently.



Verification, waiting lists and budgets are tracked by hand

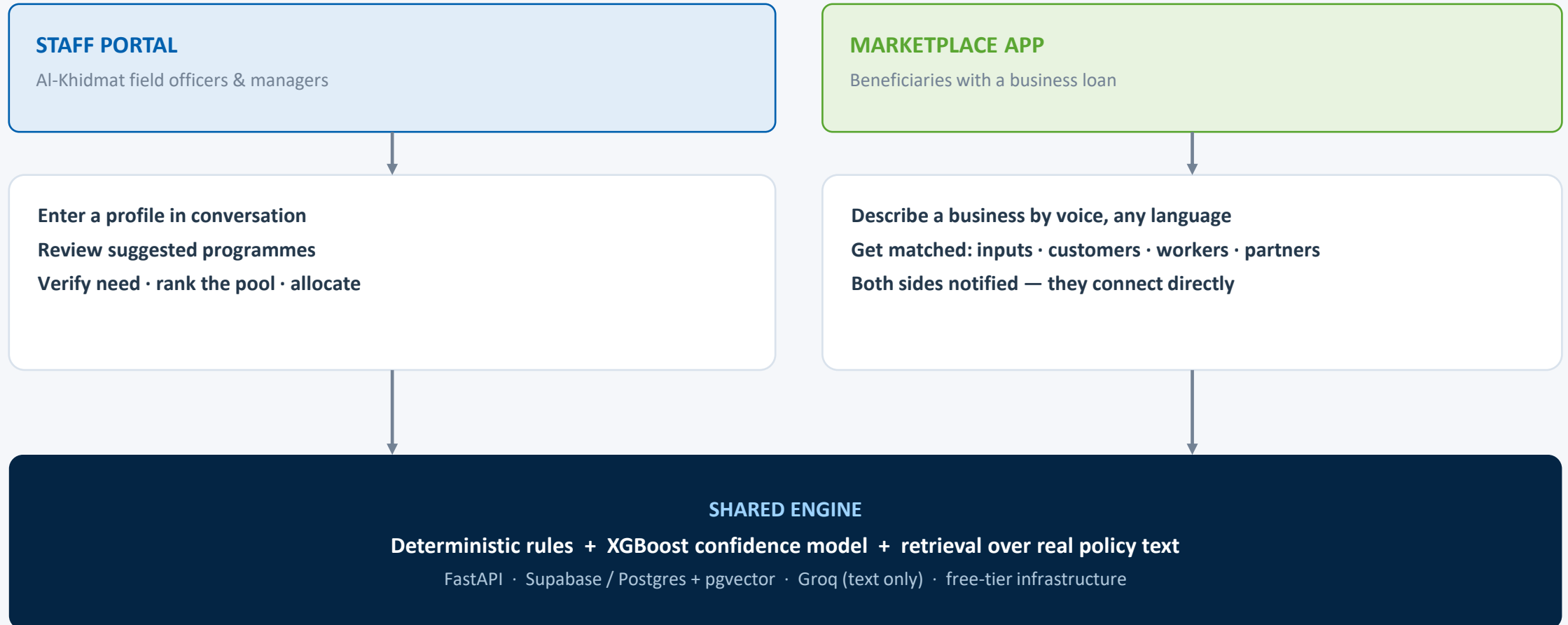
across programmes and cycles — slow, and easy to lose.



A beneficiary who starts a business is on their own

the loan ends; nobody connects them to suppliers, customers or partners.

One platform, two sides, a shared engine



Eligibility: a five-stage pipeline, humans in control



Rules ELIMINATE. The model only RANKS confidence that verification will confirm need.

No LLM anywhere in scoring · no automatic enrolment · a failed verification simply exits

Every ranking is explainable — to the factor

THE RUBRIC, SHOWN TO STAFF · one example

Income (inverse)	30%
Dependents	20%
Disability	15%
No prior assistance	20%
School-age children	15%

Weights are set per programme and snapshotted with every cycle.

STRUCTURAL FAIRNESS

How someone entered — staff-identified or walk-in — is recorded for audit and can never enter a ranking calculation.

PLAIN-LANGUAGE REASONS

Every suggestion and every rank comes with a sentence a caseworker can read aloud — traceable to the real policy text it came from.

One measurable job, done carefully

WHAT THE MODEL IS

XGBoost — gradient-boosted trees

Trained on 15,000 synthetic profiles

~54,000 labelled (profile, programme) rows

Held-out ROC-AUC ~0.76 · F1 ~0.71

Predicts one thing: will verification confirm need?

WHAT IT DOES NOT DO

No LLM in the scoring path — deterministic

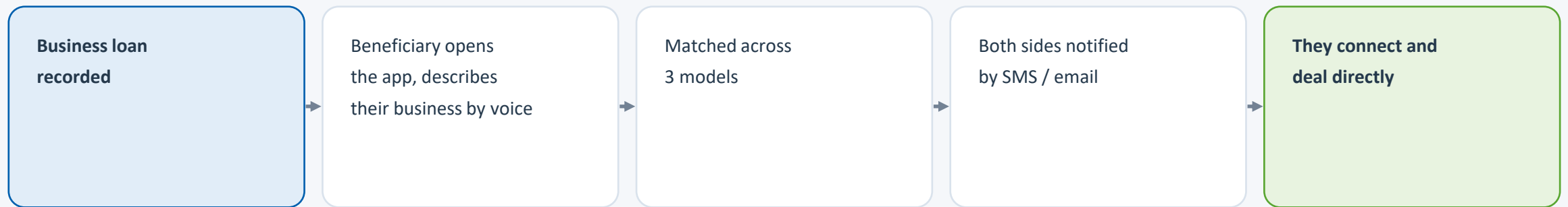
No training on real beneficiary data

Never enrolls anyone — staff decide

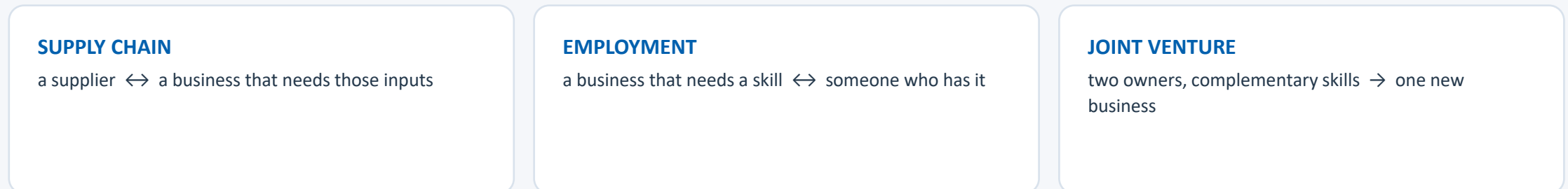
Never a funding priority — that's the rubric

Confidence shown only as low / medium / high

The marketplace: from a loan to a network



MATCHED ACROSS 3 MODELS



No fees, ever. No commission. AI-Khidmat introduces only — terms and pricing are between the two businesses.

The staff portal and the marketplace app

Two web apps over one API and engine. Every screen here is running on synthetic demo data.

STAFF PORTAL · overview

Mustahiq AI
BENEFICIARY CARE PLATFORM

Alkhidmat Foundation Pakistan · Serving humanity since 1990

SYNTHETIC DATA - LOCAL DEMO

Demo workspace | Rayan Super Admin

YOUR WORKSPACE

- Overview
- Beneficiaries
- Discovery review 275
- Outreach & verification 141
- Ranking & allocation

MANAGEMENT & INSIGHTS

- Programs & criteria
- Duplicate review 10
- Department reports
- Staff assistant

Staff portal / Overview

Demonstration workspace. People, budgets, and program policies are fictional.

OVERVIEW

Welcome back. 7 Sept 2026 [+ Register beneficiary](#)

What needs your team's attention today.

- 183** Registered beneficiaries
Across your accessible casework
- 275** Suggestions to review
Discovery matches awaiting a staff decision
- 141** Awaiting outreach
Ready for a verification visit
- 207** Verified candidates
In the shared candidate pool

Your next steps
Where cases are waiting on a staff action. [View worklist](#)

- Review discovery matches**
275 suggestions need a staff decision →
- Reach out and verify need**
141 cases are ready for assessment →
- Review the candidate pool**
207 verified candidates across programs →

Discovery is a suggestion. Verification comes before allocation.

Pipeline
Where each case sits, from registration to support.

Step	Status	Count
1	Registered	183
2	Awaiting review	275
3	In outreach	141
4	Verified pool	207
5	Supported	0

MARKETPLACE APP

Mustahiq AI Marketplace مستحق بازار

Test User 5332

District: Lahore

Trade category: Trading businesses

[+ Create a Listing](#) فہرست بنائیں

Filters & search فیلٹر

Browsing all listings (20) -- newest first

- Fahad's Grocery**
retailer Grocery / Karyana Lahore
چاہیے کیشینر اسٹور مینجر
- (unnamed business)**
service Education & Tutoring Lahore · remote-capable
آن لائن ٹیوشن
- Amina & Zainab Combined Leather Goods**
producer Manufacturing Lahore
سلائی اور چمڑے کی مشترکہ دکان
- Fahad's Grocery**
retailer Grocery / Karyana Lahore
چاہیے کیشینر اسٹور مینجر

Built end to end and tested — 41 eligibility tests, 33 staff-workflow tests, plus marketplace smoke tests.

Built for the real constraints

Free tier only

No GPU. Local CPU embeddings, hosted rules + trees, Groq free tier for text only.

Every answer is cited

Retrieval runs over uploaded policy documents — never the open web.

Any language in, English pipeline

A beneficiary speaks Urdu or Punjabi; matching still works.

Staff-operated by design

Beneficiaries never log into the eligibility side. The marketplace is the deliberate exception.

Two scheduled jobs

Bi-weekly ranking cycle and a daily marketplace cleanup — the only things not real-time.

Rate-limit aware

One Groq wrapper, live token logging, automatic fallback key on a 429.

WHAT CHANGES FOR AL-KHIDMAT

Faster, consistent, defensible aid — and a path to self-reliance

BEFORE

One programme checked per visit

Prioritisation by memory & spreadsheet

“Why them?” — no record

Loan ends, beneficiary alone

WITH MUSTAHIQ AI

Every active programme, in milliseconds

One rubric, snapshotted, factor-by-factor

A plain-language reason for every decision

Matched to a network that keeps their business going

Roadmap: real verification data → model retraining loop · row-level security · production auth & payments

From aid, to auditable aid, to self-reliance.

Mustahiq AI · Alibaba Cloud AI Hackathon · hosted by AI-Khidmat Foundation

Repository: github.com/zobiahussain/mustahiq-ai

Thank you.